



The University of Texas at Austin
McCombs School of Business

Enhancing Customer Retention Through Predictive Analytics

Credit-card churn classification with tuned boosting ensembles

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Advanced Machine Learning · March 2024

Post Graduate Program in AI & ML: Business Applications

McCombs School of Business, The University of Texas at Austin

PROJECT

03

Advanced Machine Learning

Ensemble classifiers (AdaBoost, gradient boosting, XGBoost, random forest) tuned across original, undersampled and SMOTE-oversampled data to flag credit-card customers at risk of attrition.

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Executive summary

97.5%

recall on held-out test set —
AdaBoost, undersampled training
data

94.1%

test accuracy

73.9%

test precision — acceptable cost of
catching nearly every attriter

16%

of 10,127 customers attrited (1,627)

Key findings

Transaction behavior predicts attrition. Total transaction amount and count are the two strongest signals; falling quarter-on-quarter activity is the third.

The model is built to miss almost no one. Recall was the primary objective: the cost of losing a customer outweighs the cost of an unnecessary retention offer.

Binned feature engineering hurt. Six models trained on binned features underperformed their unbinned counterparts and were dropped.

Strategic recommendations

01

Monitor transactional change

Flag customers whose transaction count or amount falls quarter-on-quarter, and intervene before the decline becomes attrition.

02

Tie rewards to volume

Expand reward programs that scale with transaction volume — tangible benefit for increased card use.

03

Deepen product engagement

Customers holding more bank products attrite less. Cross-sell selectively to widen each relationship.

04

Deploy retention analytics

Run the model on the live book to target retention effort at the customers most likely to leave.

05

Equip frontline staff

Surface the risk score and its drivers to customer-facing staff so early disengagement is met with a conversation.

06

Watch revolving balance

Very low and very high revolving balances both associate with attrition; the middle band is the stable one.

Business problem and solution approach

Problem

- The bank is losing credit-card customers. Use predictive analytics to learn what behaviors signal which customers are about to leave, early enough to act.

Primary metric: recall. A missed attriter is a lost customer; a false alarm is a cheap retention offer.

Approach

- EDA — univariate and bivariate against the attrition flag
- Preprocessing: imputation, encoding, 80/15/5 train-test-validation split
- Class imbalance: original, random undersampling, SMOTE oversampling
- Six base models × three sampling strategies; feature-engineered (binned) variants
- Hyperparameter tuning by randomized search; validation on held-out set
- Final test and feature importance

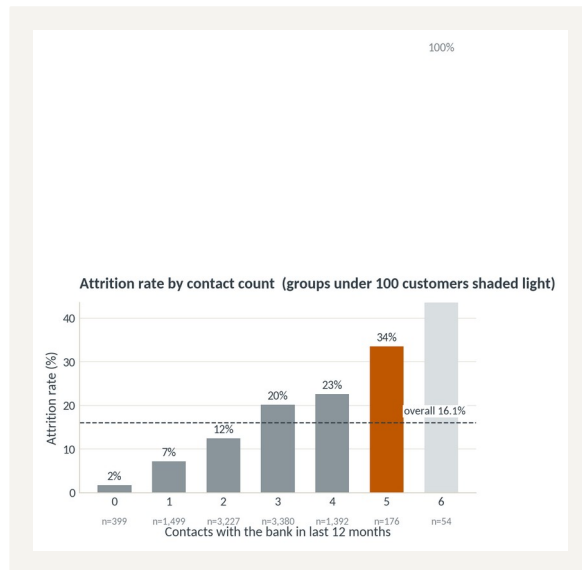
EDA results

Attrition against engagement, activity and credit behavior

Engagement and inactivity

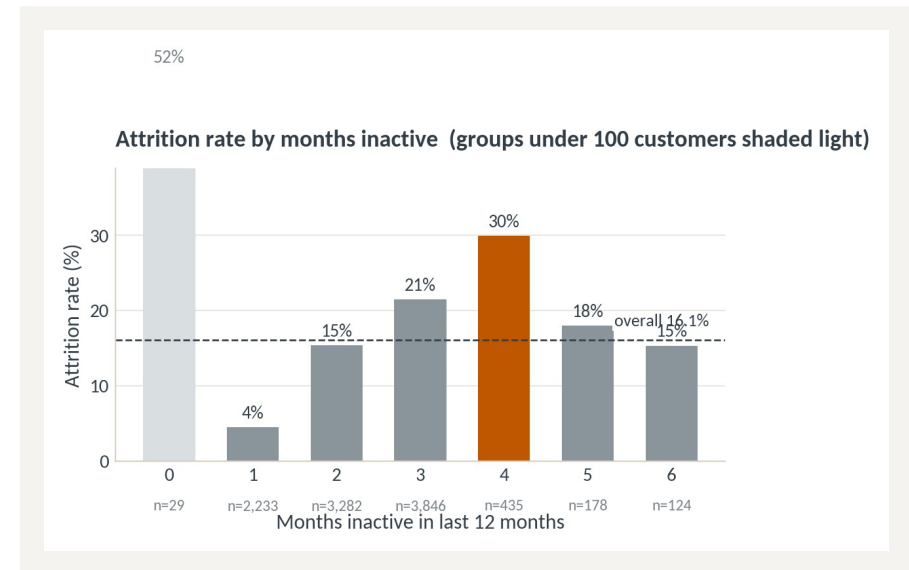
Attrition vs contact count

- Attrition rises steadily with contact count: 2% at zero contacts, 34% at five. Contact frequency reads as friction rather than engagement.



Attrition vs months inactive

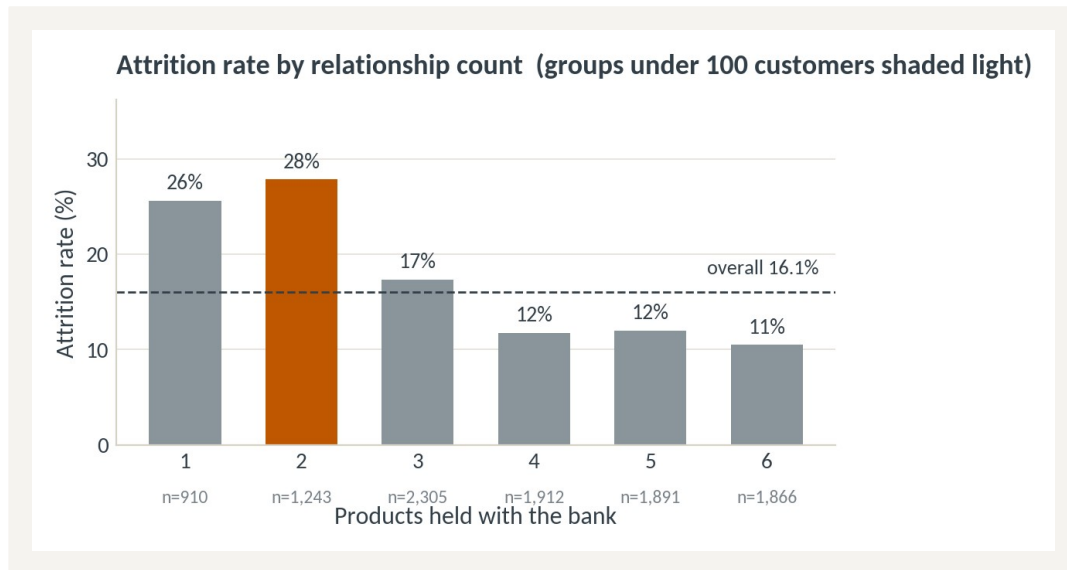
- Two to three inactive months is common across the book; attrition climbs from 4% at one month to 30% at four.



Relationship breadth and revolving balance

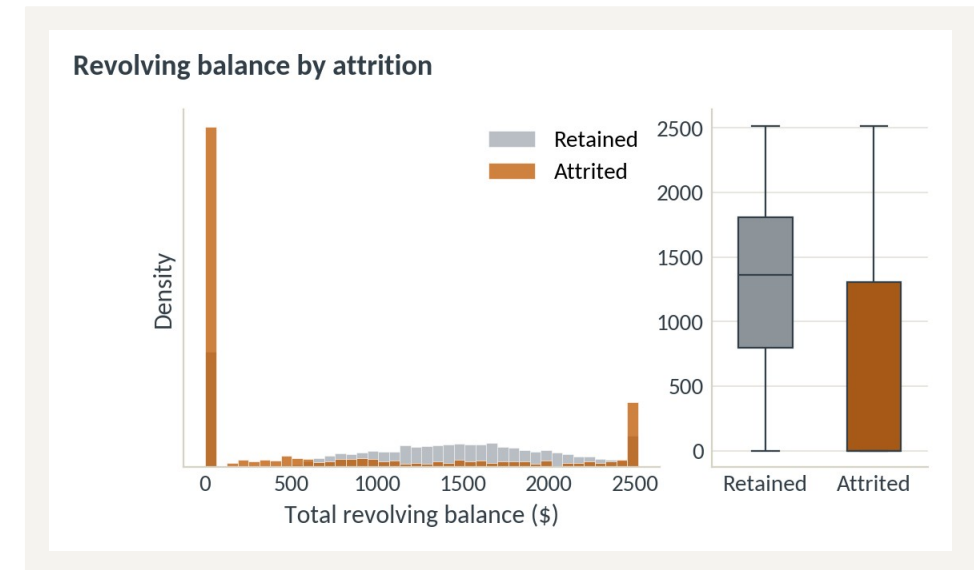
Attrition vs total relationship count

- More products, less attrition. Customers with one or two products leave at 26–28%; those with four or more at 11–12%.



Attrition vs total revolving balance

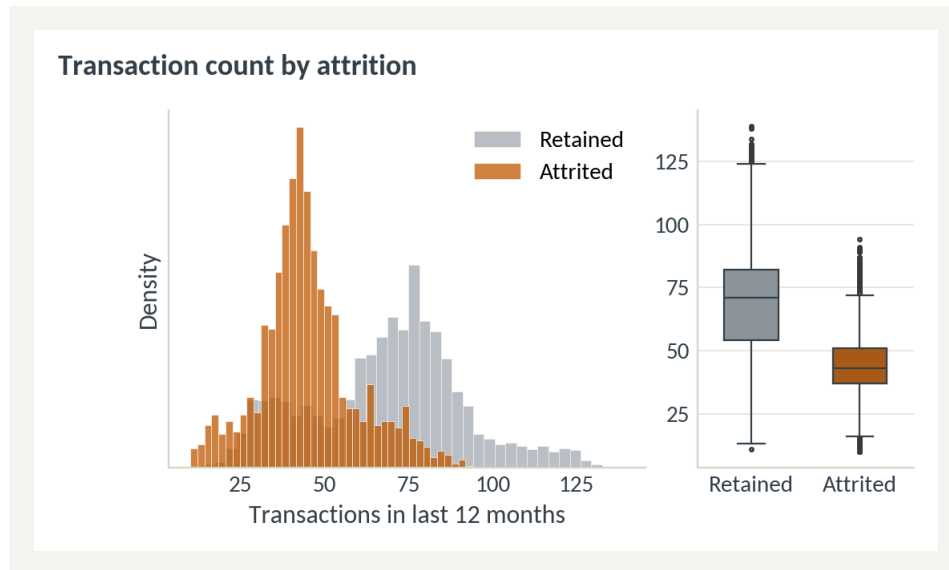
- Attriters cluster at zero revolving balance (median \$0 against \$1,364 for retained); a moderate balance is associated with staying.



Transaction activity

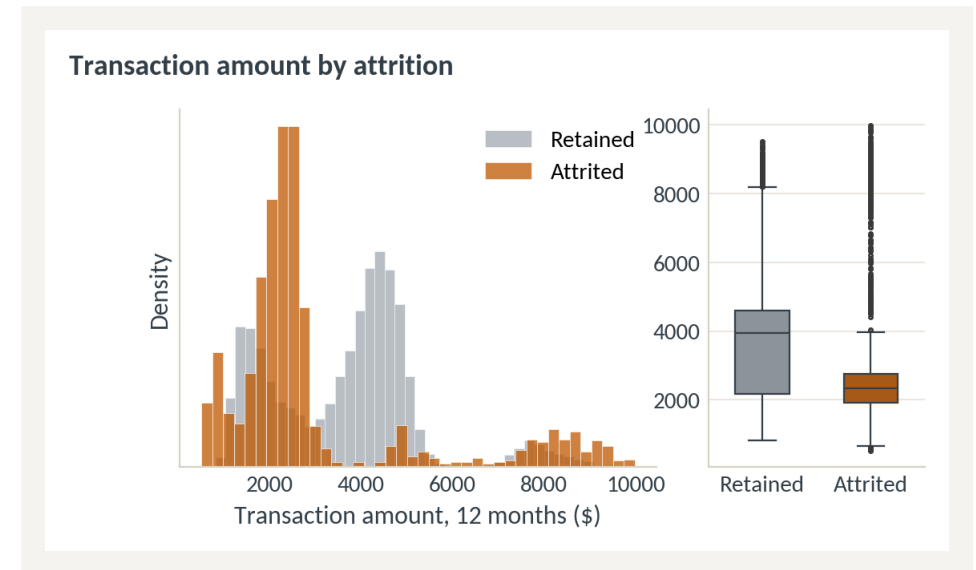
Attrition vs transaction count

- Attriters transact far less often. The two distributions barely overlap above 80 transactions a year.



Attrition vs transaction amount

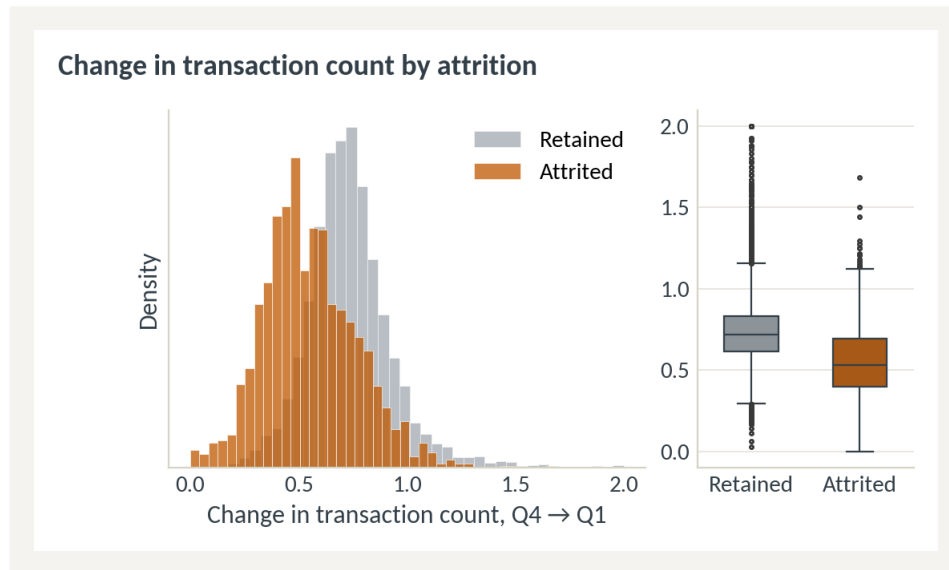
- Lower annual spend accompanies attrition; the retained group carries the long upper tail.



Change in behavior, Q4 to Q1

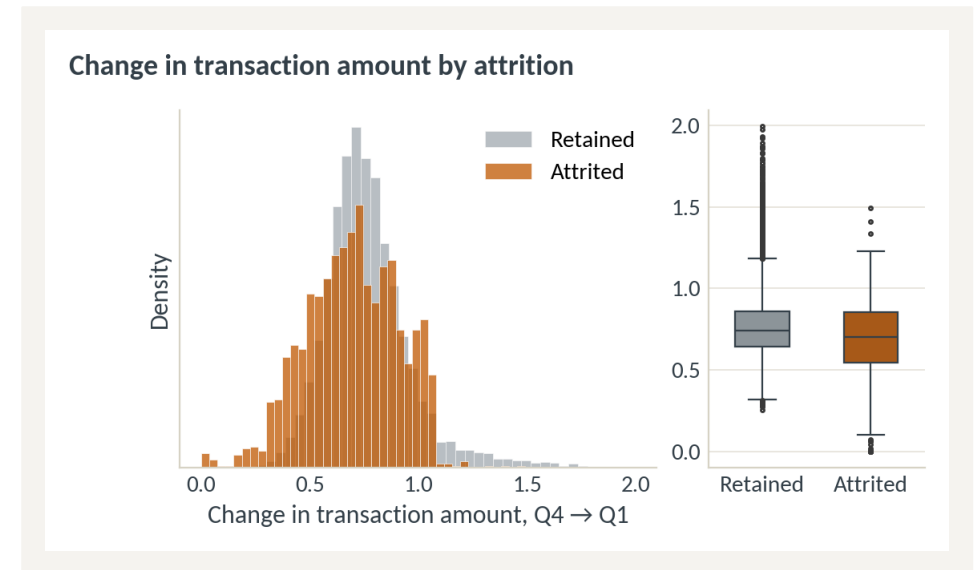
Change in transaction count

- Median change is higher for retained customers: those who hold or increase transaction frequency stay.

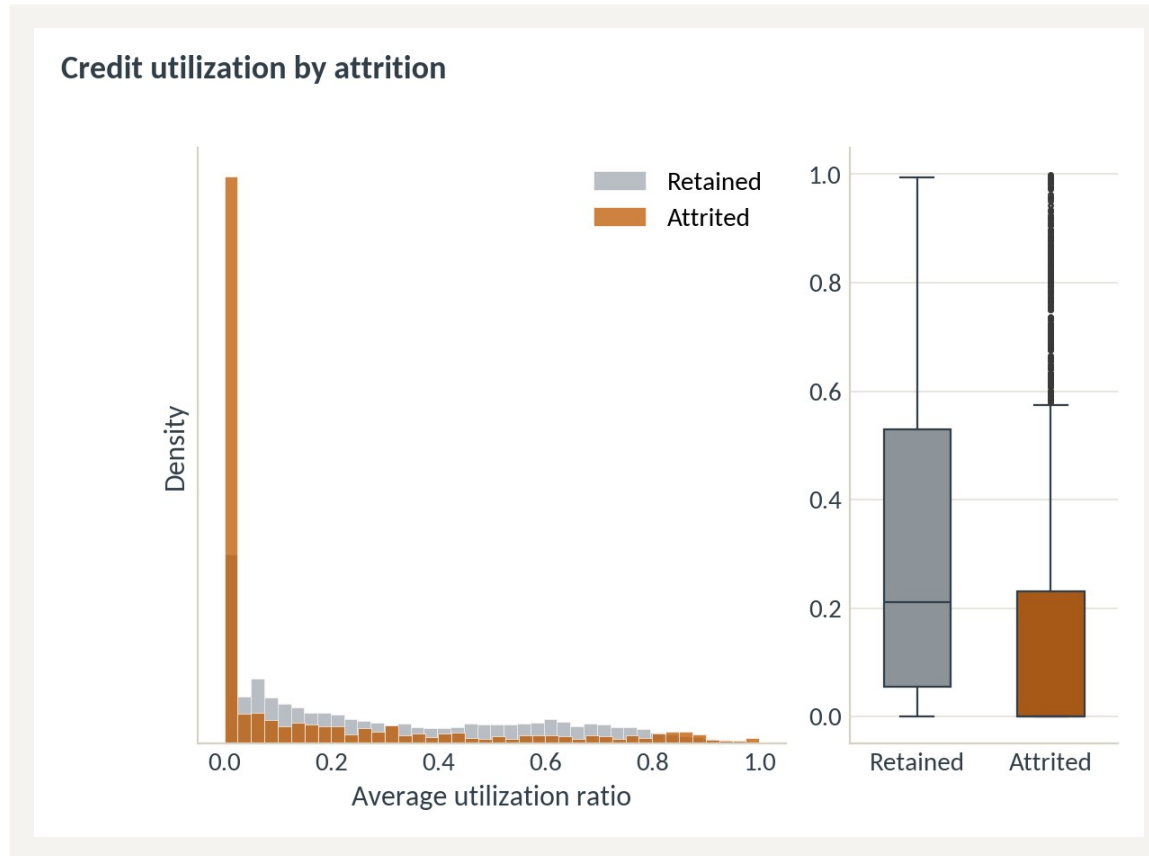


Change in transaction amount

- Retained customers spend consistently quarter to quarter; attriters show wider variance in the change.



Credit utilization



- Low utilization ratio is far more common among attriters.
- The retained group spreads across the full range, with fewer customers at the low end.
- Consistent with the transaction findings: attriters are customers who had already stopped using the card.

Data preprocessing

Cleaning and imputation

Duplicates. None.

Missing values. Education level (1,519) and marital status (749), imputed by most frequent value inside the pipeline.

Outliers. Present but plausible; no treatment.

Encoding. One-hot encoding of categorical features.

Split and sampling

Training 80% of the data; of the remaining 20%, 75% to test (15% overall) and 25% to validation (5% overall).

Three training sets: original, random undersampling of the majority class, SMOTE oversampling of the minority class ($k = 5$, 1:1).

Feature engineering. A second run binned months on book, credit limit, open-to-buy, revolving balance, transaction amount and count, utilization and card category. Binned models underperformed and were dropped.

Tuning. Randomized search over AdaBoost, gradient boosting, XGBoost and random forest.

Model performance and selection

Base models on undersampled data

Six untuned classifiers, recall on training and validation sets. Boosting and random forest generalize; the decision tree and bagging overfit.

Model	Training recall	Validation recall
Bagging	0.994	0.926
Random forest	1.000	0.963
Gradient boosting	0.982	0.938
AdaBoost	0.952	0.951
XGBoost	1.000	0.975
Decision tree	1.000	0.901

XGBoost and random forest reach perfect training recall — a warning sign for overfitting that tuning addresses.

AdaBoost shows the smallest train-validation gap (0.001): the most honest baseline.

All four ensembles were carried into tuning across the three sampling strategies.

Tuned models — validation performance by sampling strategy

Model • training data	Accuracy	Recall	Precision	F1
AdaBoost • original	0.980	0.901	0.973	0.936
AdaBoost • undersampled	0.951	0.963	0.780	0.862
AdaBoost • oversampled	0.980	0.951	0.928	0.939
Gradient boosting • undersampled	0.947	0.938	0.776	0.849
Gradient boosting • original	0.970	0.864	0.946	0.903
Gradient boosting • oversampled	0.976	0.951	0.906	0.928
XGBoost • original	0.959	0.951	0.819	0.880
XGBoost • undersampled	0.576	1.000	0.274	0.430
XGBoost • oversampled	0.572	1.000	0.272	0.427

Three candidates for test: AdaBoost undersampled (recall 0.963), AdaBoost oversampled (0.951), XGBoost original (0.951).

XGBoost on resampled data collapsed: perfect recall at 27% precision means it flags nearly everyone. Discarded.

Original-data models trade recall for precision — the wrong trade for this objective.

Training-set metrics are in the appendix; the ranking is unchanged.

Test-set results and model selection

Candidate on test set	Accuracy	Recall	Precision	F1
AdaBoost • undersampled (selected)	0.941	0.975	0.739	0.841
AdaBoost • oversampled	0.962	0.934	0.844	0.887
XGBoost • original	0.950	0.955	0.782	0.860

97.5%

recall — fewest attriters missed

94.1%accuracy — stable across train,
validation and test**73.9%**precision — about one retention
offer in four is unnecessary**84.1%**

F1 — balance of the two

Selected: AdaBoost, tuned, trained on undersampled data — 75 estimators, learning rate 0.1, depth-3 trees.

Why undersampled over oversampled

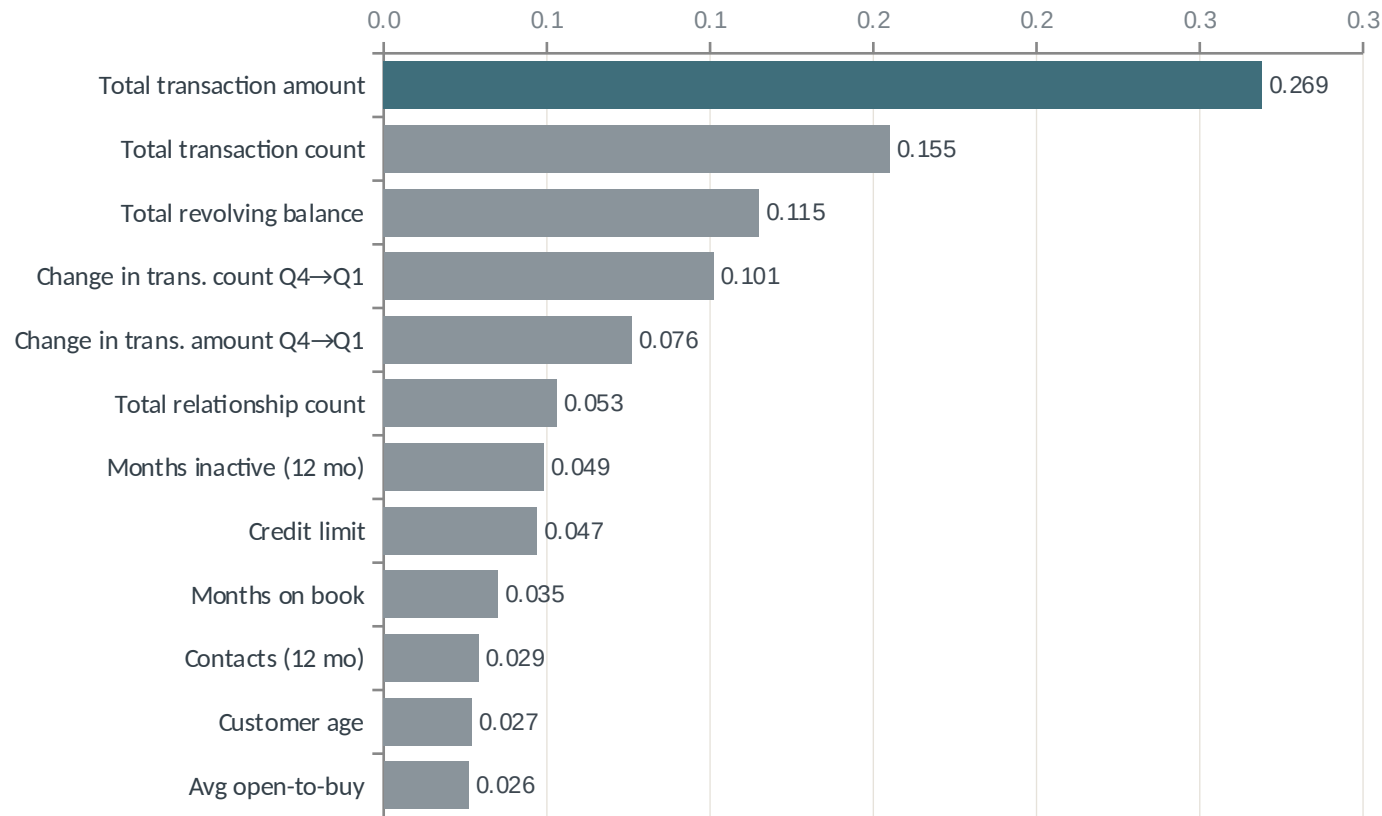
Real-world representation. Training on actual customer records only — no synthetic rows — so behavioural insights come from genuine interactions.

Generalization. Oversampling scores higher on precision and F1, but one-third of its training set is synthetic. A model fitted to real profiles should transfer better to the live book.

Stakeholder trust. Predictions that rest on verifiable customer profiles are easier to defend to the people who will act on them.

The trade. Recall is the objective and precision cannot be ignored. 97.5 / 73.9 is the right compromise for a bank that would rather over-contact than lose customers.

Feature importance — selected model



Top two: transaction amount and count (0.27, 0.16).

Customers with lower overall spend and fewer transactions are the ones who leave.

Revolving balance and quarter-on-quarter change (0.12, 0.10, 0.08). How credit is used, and how that use shifts, is the second tier of risk.

Moderate: relationship count, inactivity, credit limit. Breadth of relationship and dormancy add signal.

Marginal: tenure, contacts, age, open-to-buy. Some relevance; not decision-driving.

Demographics contribute nothing. Gender, education, income, marital status, card tier and dependants all score below 0.01.

Appendix

Sampling detail, training metrics, data background

Tuned models — training performance

Model • training data	Accuracy	Recall	Precision	F1
AdaBoost • original	0.981	0.922	0.962	0.941
AdaBoost • undersampled	0.981	0.986	0.976	0.981
AdaBoost • oversampled	0.982	0.986	0.979	0.982
Gradient boosting • undersampled	0.976	0.981	0.972	0.976
Gradient boosting • original	0.974	0.876	0.956	0.915
Gradient boosting • oversampled	0.924	0.855	0.991	0.918
XGBoost • original	0.975	0.992	0.869	0.926
XGBoost • undersampled	0.768	1.000	0.683	0.812
XGBoost • oversampled	0.758	1.000	0.673	0.805

Oversampling: SMOTE. Sampling strategy 1 (minority resampled to match majority, 1:1); k = 5 nearest neighbors; random state 1.

Undersampling: random under-sampler. Random state 1, so the same majority subset is removed on every run.

Validation set: 5% of the data (25% of the 20% hold-out); test set 15%.

Data background — customer profile

10,127 customers, 21 variables; 1,627 attrited (16.1%).

Variable	Range	Center	Reading
Customer_Age	26 – 73	mean $\approx$ median $\approx$ 46	mature customer base
Dependent_count	0 – 5	$\approx$ 2	family-oriented
Months_on_book	13 – 56	$\approx$ 36, sd $\approx$ 8	mix of new and long-term
Total_Relationship_Count	1 – 6	3 – 4	multiple products per customer
Months_Inactive_12_mon	0 – 6	$\approx$ 2, upper quartile 3	two to three inactive months typical
Contacts_Count_12_mon	0 – 6	$\approx$ 2.5, upper quartile 3	moderate engagement or support load

Data background — credit and transaction behavior

Variable	Range	Center	Reading
Credit_Limit	1,438 – 34,516	mean $\approx$ 8,632, sd $\approx$ 9k	wide range of creditworthiness
Total_Revolving_Bal	0 – 2,517	mean $\approx$ 1,162, sd $\approx$ 800	significant variability in carried debt
Avg_Open_To_Buy	3 – 34,516	mean $\approx$ 7,469, sd $\approx$ 9,090	varied spending headroom
Total_Amt_Chng_Q4_Q1	0 – 3.4	low mean, tight IQR	stable spending habits or seasonality
Total_Trans_Amt	510 – 18,484	broad spread	modest spenders and a high-spending segment
Total_Trans_Ct	10 – 139	regular use typical	diverse frequency; sparse and heavy users at the tails
Total_Ct_Chng_Q4_Q1	0 – 3.7	most values $\approx$ 1	consistent usage quarter to quarter
Avg_Utilization_Ratio	0 – 0.999	mean 0.275, sd 0.276	low utilization overall; near-zero to fully utilized



Thank you

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